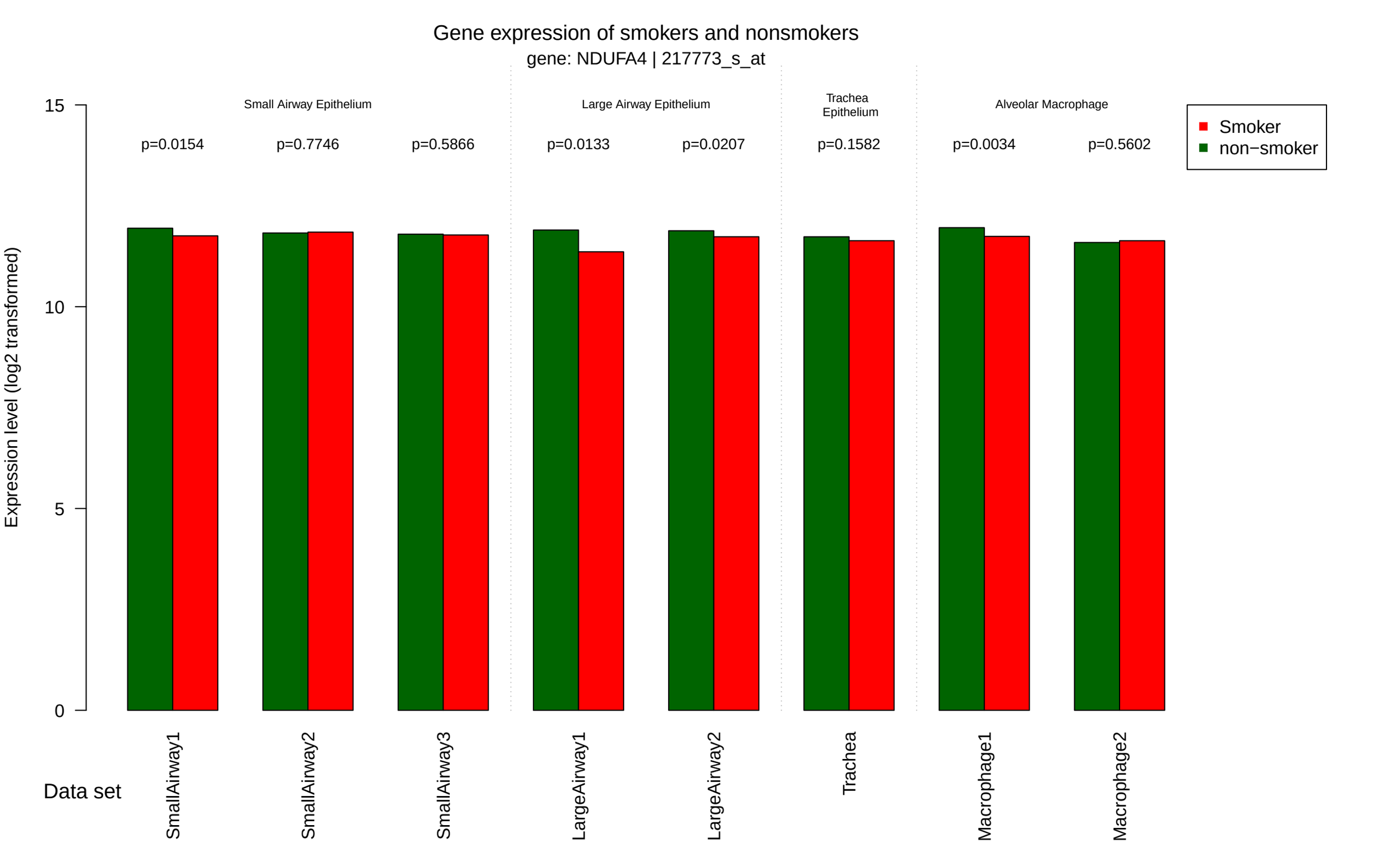
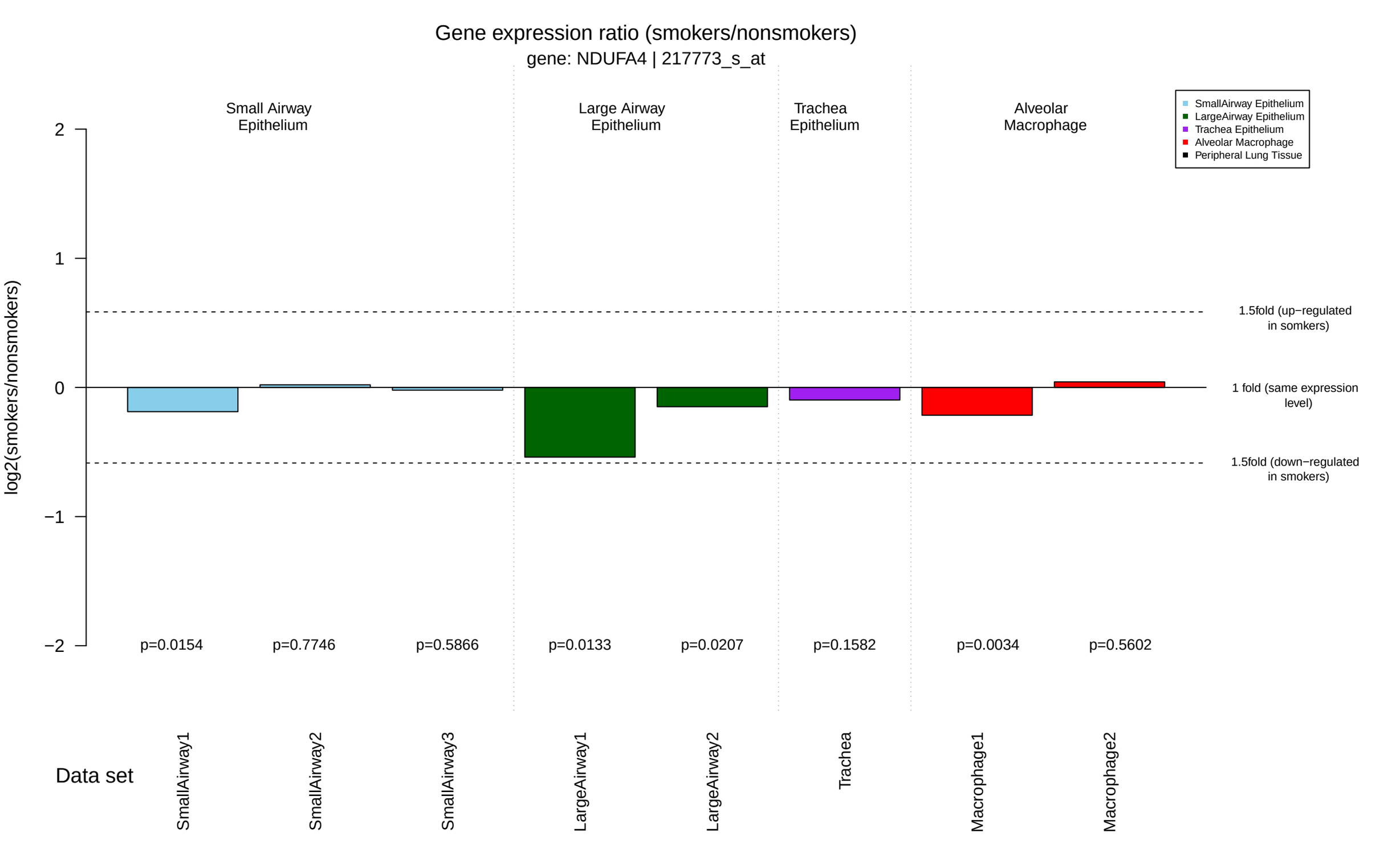
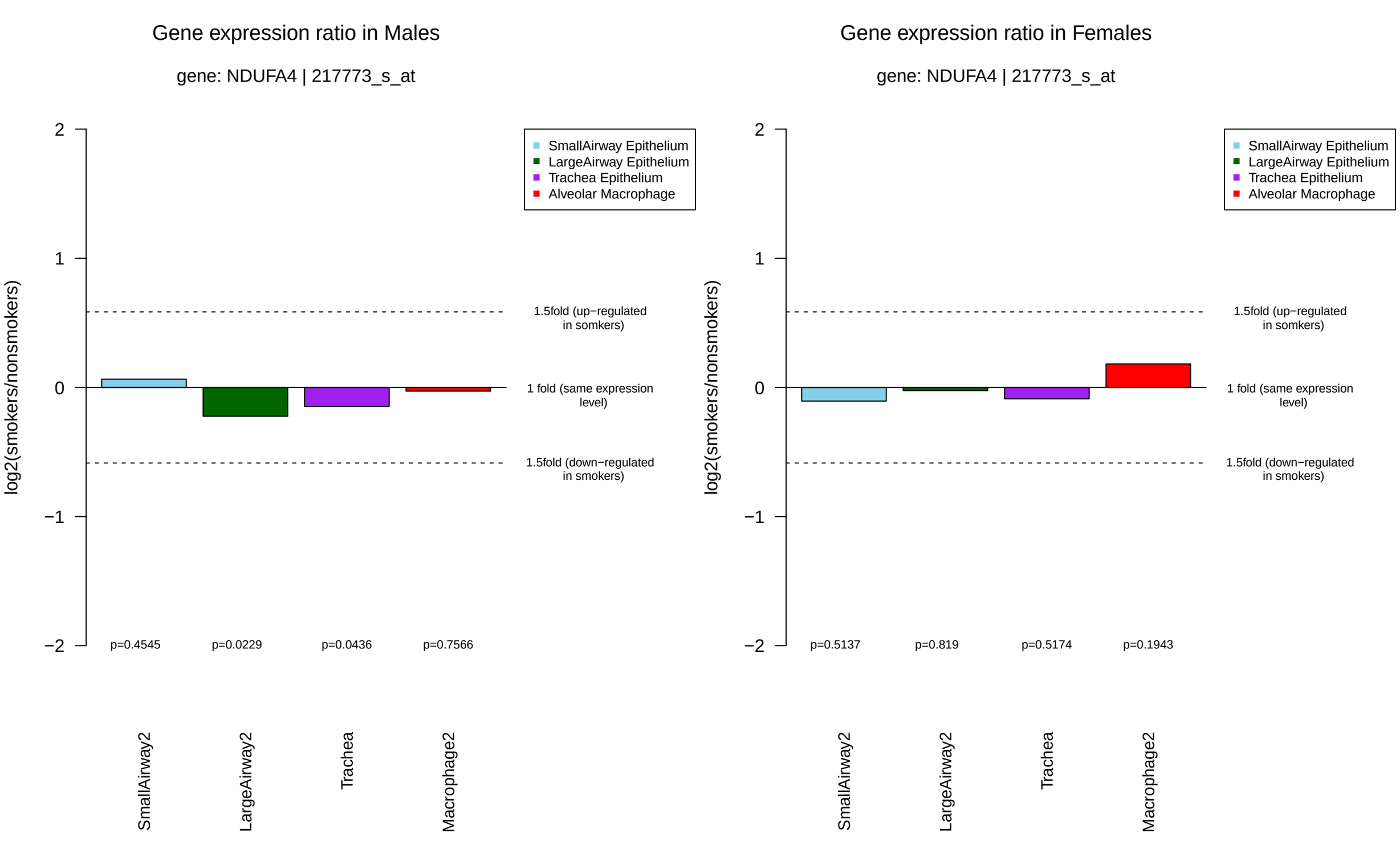
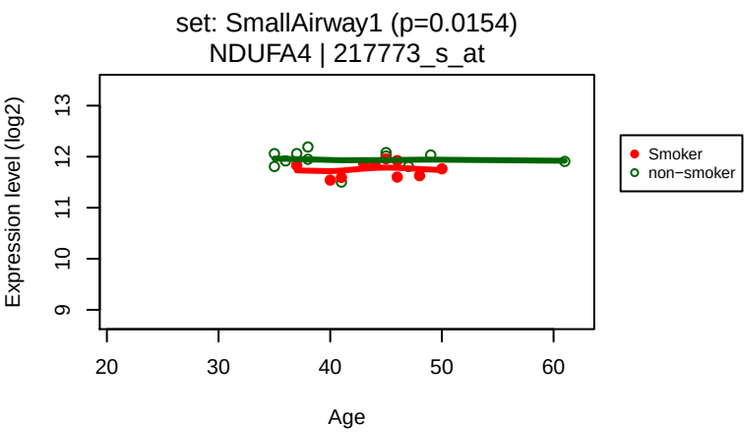
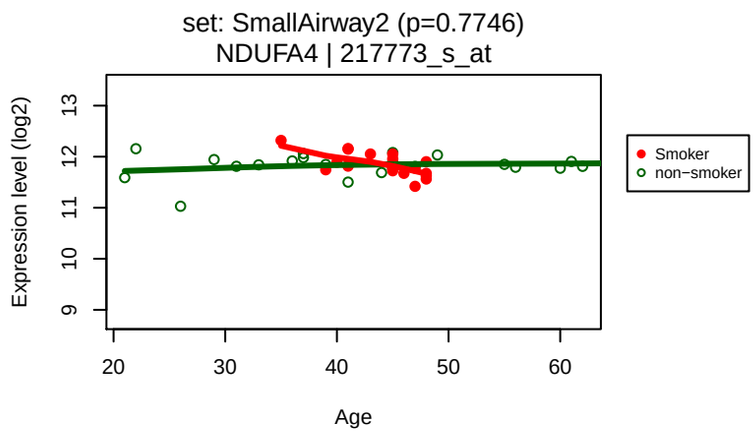
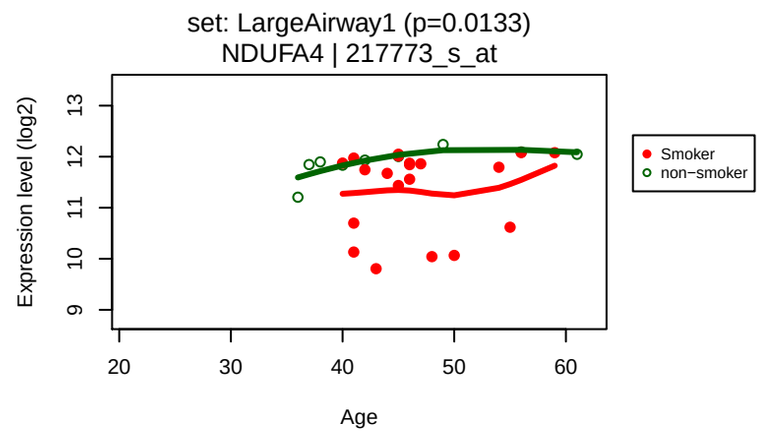

Gene expression of smokers and nonsmokers
gene: NDUFA4 | 217773_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0154
p=0.7746
p=0.5866
p=0.0133
p=0.0207
p=0.1582
p=0.0034
p=0.5602
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

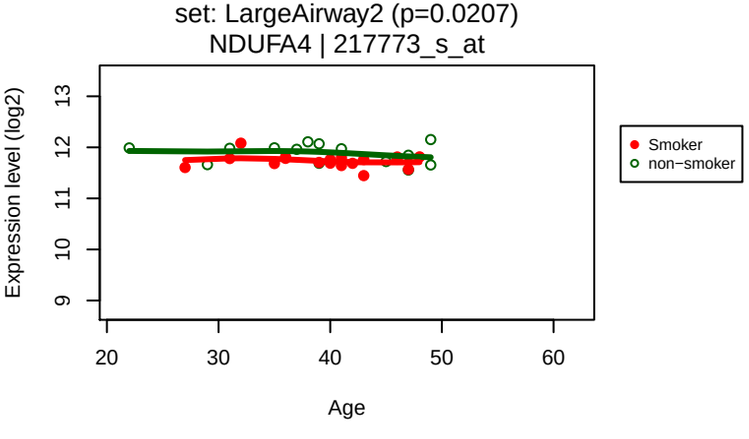

Gene expression ratio (smokers/nonsmokers)
gene: NDUFA4 | 217773_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0154
p=0.7746
p=0.5866
p=0.0133
p=0.0207
p=0.1582
p=0.0034
p=0.5602
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

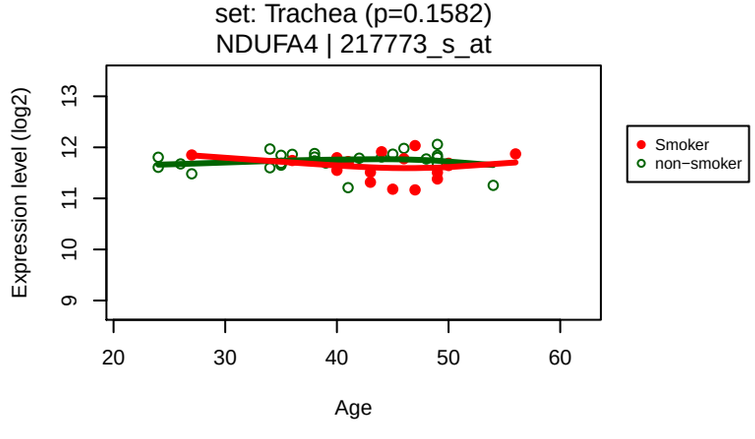

Gene expression ratio in Males
gene: NDUFA4 | 217773_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4545
p=0.0229
p=0.0436
p=0.7566
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NDUFA4 | 217773_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5137
p=0.819
p=0.5174
p=0.1943
SmallAirway2
LargeAirway2
Trachea
Macrophage2

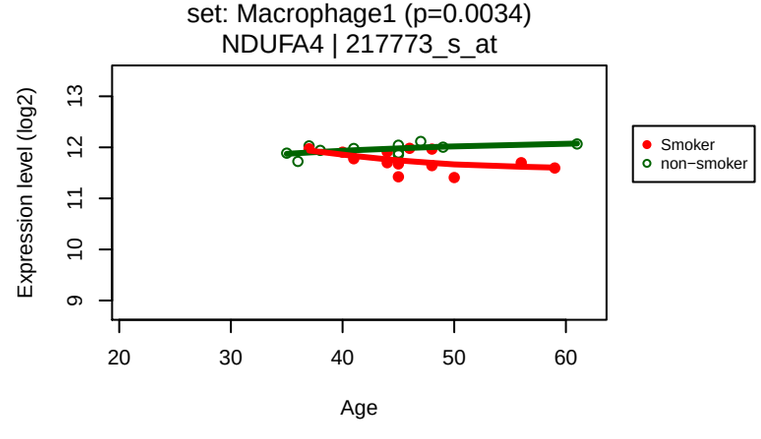

set: SmallAirway1 (p=0.0154)
NDUFA4 | 217773_s_at
Expression level (log2)
Age
Smoker
non−smoker

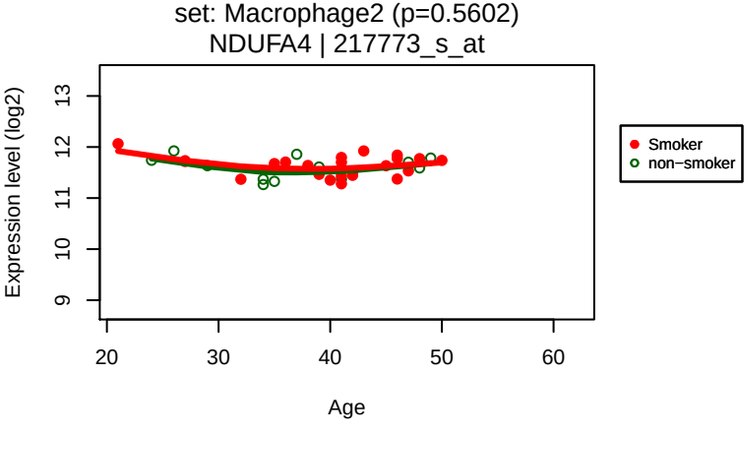

set: SmallAirway2 (p=0.7746)
NDUFA4 | 217773_s_at
Expression level (log2)
Age
Smoker
non−smoker

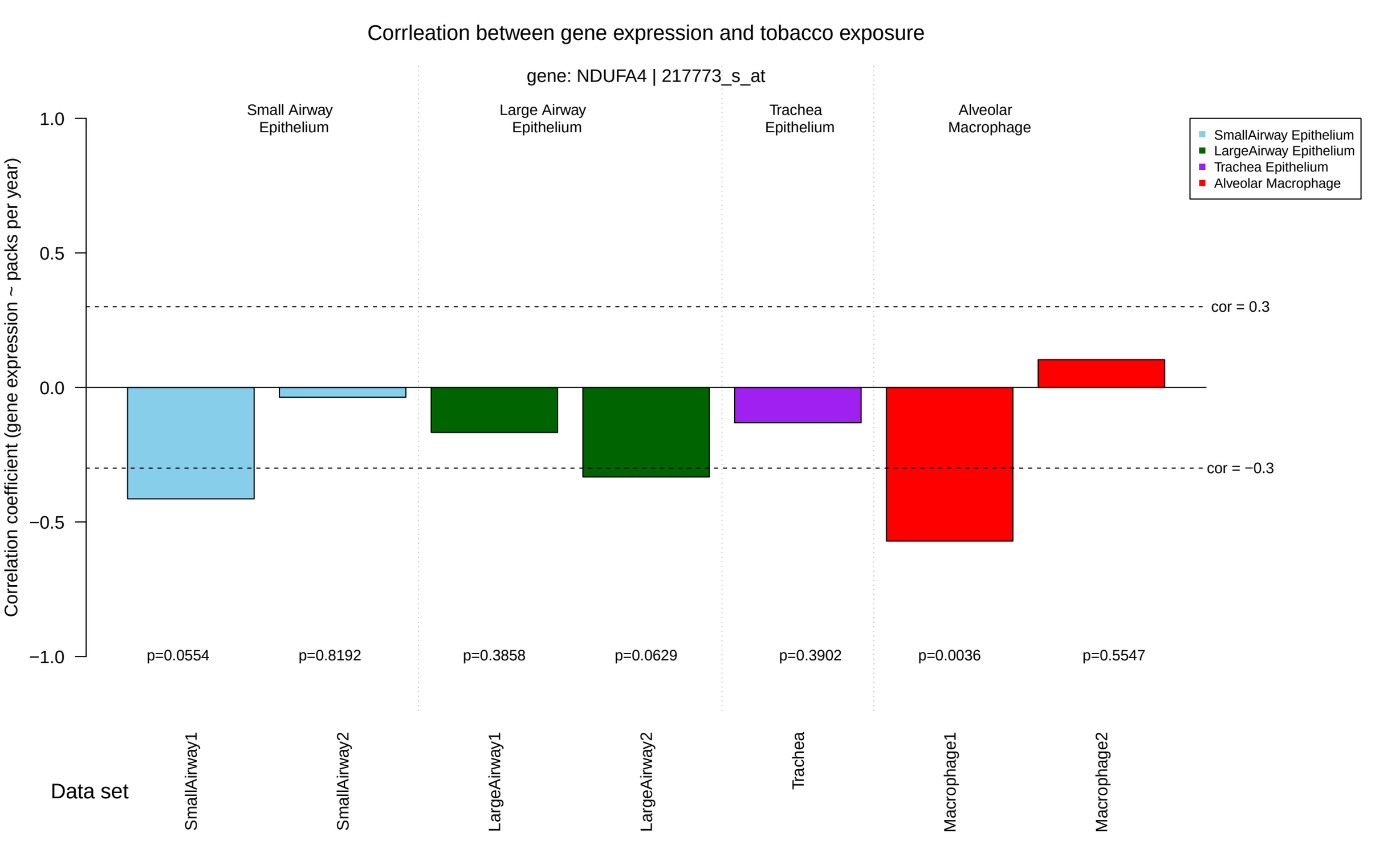

set: LargeAirway1 (p=0.0133)
NDUFA4 | 217773_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0207)
NDUFA4 | 217773_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1582)
NDUFA4 | 217773_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0034)
NDUFA4 | 217773_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5602)
NDUFA4 | 217773_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NDUFA4 | 217773_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0554
p=0.8192
p=0.3858
p=0.0629
p=0.3902
p=0.0036
p=0.5547
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2